# Variable-rate streaming with a spike-count readout

exp082 · 10 August 2026 · Draft


## Abstract

This experiment tests whether PING networks trained across input rates can classify continuous MNIST streams with the same spiking readout used during training. Excitatory spikes drive ten output LIF neurons, and each class logit is the corresponding neuron's total spike count within the current digit's presentation window. A variable-rate, variable-duration stream shows the online network response, while a 200 ms rate psychometric and a duration-by-rate map measure performance across the trained input distribution.


## Methods

### 1. Training source

The experiment loads `ping__variable_rate__seed42`, `ping__variable_rate__seed43`, and `ping__variable_rate__seed44` from exp022. Each PING network is trained by sampling one maximum-pixel Poisson rate independently per image presentation from 0.5, 0.75, 1, 1.5, 2, 3, 5, 7.5, 10, 15, and 25 Hz. The recurrent weights and classifier are frozen throughout this experiment.

**Note.** The exp022 cells are registered and `tools/snn` implements the required output-LIF `spike-count` training, restore, and output-raster path. Presentation boundaries reset only the output LIF and its counter; the hidden PING state remains continuous. This entry still fails loudly if the variable-rate checkpoint bank is absent or uses another readout.

Streaming inference uses each seed's checkpoint selected by validation accuracy, because this is a deployment evaluation of the selected classifier rather than an analysis of final-epoch drift.

### 2. Spike-count readout

At each timestep, the 1024-element excitatory spike vector $\boldsymbol{s}^{E(t)}$ drives a trained 1024-by-10 projection $W_{\text{out}}$ into ten output LIF neurons. For a presentation beginning at timestep $a$ and ending before timestep $b$, the cumulative spike-count evidence for class $c$ at timestep $u$, where $a \leq u < b$, is

$$z_{c(u)} = \sum_{t=a}^{u} s_c^{\text{out}(t)}, \quad c \in \{0, ..., 9\}. \quad (1)$$

Here $s_c^{\text{out}(t)}$ is the spike emitted by output LIF neuron $c$ at timestep $t$, and $\boldsymbol{z}(u)$ contains the ten dimensionless cumulative output-spike counts. The final spike-count logit is $z_{c(b-1)}$, and the predicted digit is $\arg\max_c z_{c(b-1)}$. Dividing final counts by the common presentation duration gives firing rates for reporting and leaves this argmax unchanged, but rates are not passed to cross-entropy as logits. The output neurons therefore have membrane state, leak, threshold, spikes, and reset; unlike `mem-mean`, their membrane voltages are not the logits.

**3. Single-trial evidence visualization**

Before the continuous stream, one matched-condition trial is shown in isolation. The first correctly classified presentation in the pre-existing matched stream is selected for this explanatory figure: digit 4, presented for 200 ms at a maximum-pixel rate of 5 Hz. The selection is deliberately outcome-based so the figure can explain a successful readout trajectory; it is not used to estimate performance. The output-LIF state and spike counter reset at the presentation boundary, while the hidden PING state continues from the preceding presentation. The displayed class trace is

$$p_{c(u)} = \mathrm{softmax}_{c(\boldsymbol{z}(u))}. \quad (2)$$

Here $p_{c(u)}$ is the softmax visualization of cumulative integer output-spike counts for class $c$. It is not a calibrated posterior probability. The trial is used to explain how discrete output spikes produce the class-evidence traces; aggregate performance is reported separately.

A second view enlarges 91.5–94.5 ms of the same stored trial. This interval is selected post hoc to explain a conspicuous transition in the full-trial trace, not to estimate performance. It aligns individual output-LIF spikes with the corresponding cumulative class counts and their softmax transformation.

**4. Variable-rate and variable-duration inference**

Every digit is read from only the spikes generated during its own presentation:

$$T_{\mathrm{readout}} = T_{\mathrm{presentation}}. \quad (3)$$

Here $T_{\mathrm{readout}}$ is the spike-summation window and $T_{\mathrm{presentation}}$ is the duration of the corresponding digit. The five-digit headline stream changes both presentation duration and maximum-pixel Poisson rate at digit boundaries. A factorial evaluation then crosses presentation durations 25, 50, 100, and 200 ms with the eleven training rates. Every cell is evaluated on 200 classified digits independently for seeds 42, 43, and 44. Independent five-digit streams are evaluated in simulator batches; each batch member has separate neuronal state, while hidden state remains continuous only between the five digits belonging to that stream. Statistical grid cells retain compact per-presentation spike counts and population totals. Full rasters are retained for the illustrative stream and an internal matched-condition validation stream.

**5. Psychometric protocol**

The rate psychometric fixes presentation and readout duration at 200 ms, varies the maximum-pixel rate across the eleven training values, and reports held-out classification accuracy across the three trained seeds. This isolates rate robustness from shortened evidence windows.

## Results

### 1. Single-trial spike-count evidence

The isolated trial shows how the network turns successive bursts of activity into a decision. Excitatory spikes drive the ten output-LIF neurons, one for each class. Every output spike adds one to that class's cumulative count. These counts can only increase, remain unchanged between spikes, and determine the final prediction directly: whichever class has accumulated the most spikes wins.

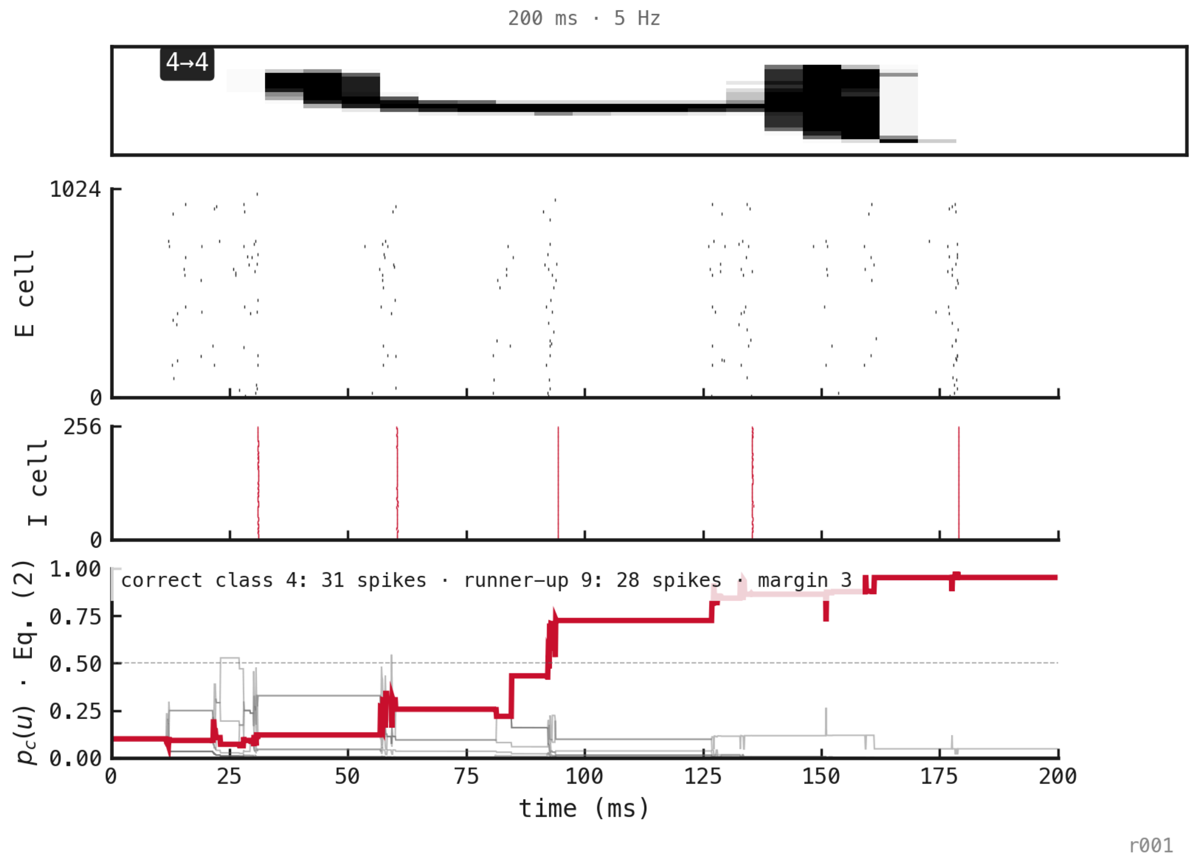


Figure 1: A correctly classified 200 ms presentation at a maximum-pixel input rate of 5 Hz, selected as the first success in the pre-existing matched stream. The output readout resets at the boundary while hidden PING state remains continuous. The top badge gives the true label and final prediction. The middle panels show E- and I-cell spike rasters. The lower panel shows $p_{c(u)}$ from Equation 2; the true and winning class is red, and the annotation gives its final count, the runner-up count, and the winning margin. Equation 2 visualizes the cumulative integer counts $z_{c(u)}$ from Equation 1 and is not a calibrated posterior probability.

The plotted traces apply softmax to these cumulative counts. Softmax does not measure how much evidence exists in absolute terms; it shows how that evidence is distributed among the competing classes. Adding the same number of spikes to every class leaves the traces unchanged. What matters is the difference between their counts. For any two classes $c$ and $d$,

$$\frac{p_{c(u)}}{p_{d(u)}} = \exp\Big(z_{c(u)} - z_{d(u)}\Big).$$

Their displayed evidence ratio is therefore determined entirely by their spike-count margin. Each additional spike increases that class's log-odds against every unchanged competitor by one. Consequently, a modest accumulated lead can appear as near-total confidence even when the underlying difference is only a few spikes. The traces show relative, exponentially emphasized evidence, not calibrated probabilities that a class is correct.

This also explains why other classes can rise above the eventual winner during the first few cycles. With only a handful of output spikes, a single event can change the leader. The output counts reset for the new digit, but the hidden PING state continues from the preceding presentation, so the earliest activity need not favour the new input cleanly. Class 4 wins only after its evidence accumulates more consistently across later bursts.

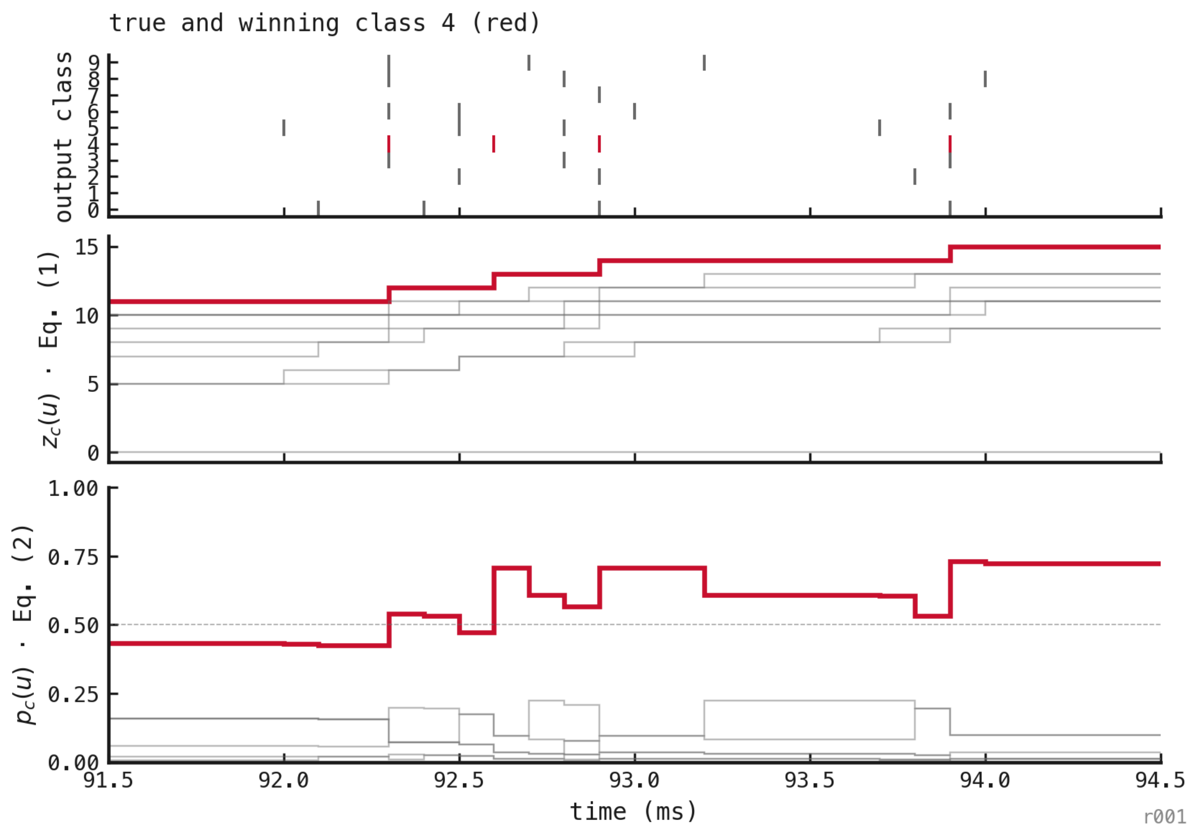


Figure 2: The 91.5–94.5 ms transition from Figure 1. Top: output-LIF spikes by class. Middle: cumulative counts $z_{c(u)}$ from Equation 1. Bottom: $p_{c(u)}$ from Equation 2. The true and winning class 4 is red; other classes are grey. The apparent jump in class evidence is produced by discrete count increments rather than by a continuous-valued readout.

Figure 2 resolves how the rhythmic E/I activity becomes the plotted class evidence. Output spikes occur around the population bursts, and each mark in the top panel increments one cumulative count in the middle panel. When one class spikes, its softmax trace rises and all the others fall because they share the same normalization denominator. A class's displayed evidence can therefore decrease even while its own cumulative count remains unchanged. The connecting lines render samples at successive timesteps; the underlying readout changes discretely at output spikes. This trial is consistent with successive PING bursts delivering packets of class evidence, but it does not establish that every cycle independently improves the decision.

**2. Streaming classification and temporal evidence**

The variable stream changes both input rate and presentation duration at digit boundaries. Each decision still uses exactly the spikes emitted during that digit, so evidence never leaks across labels.

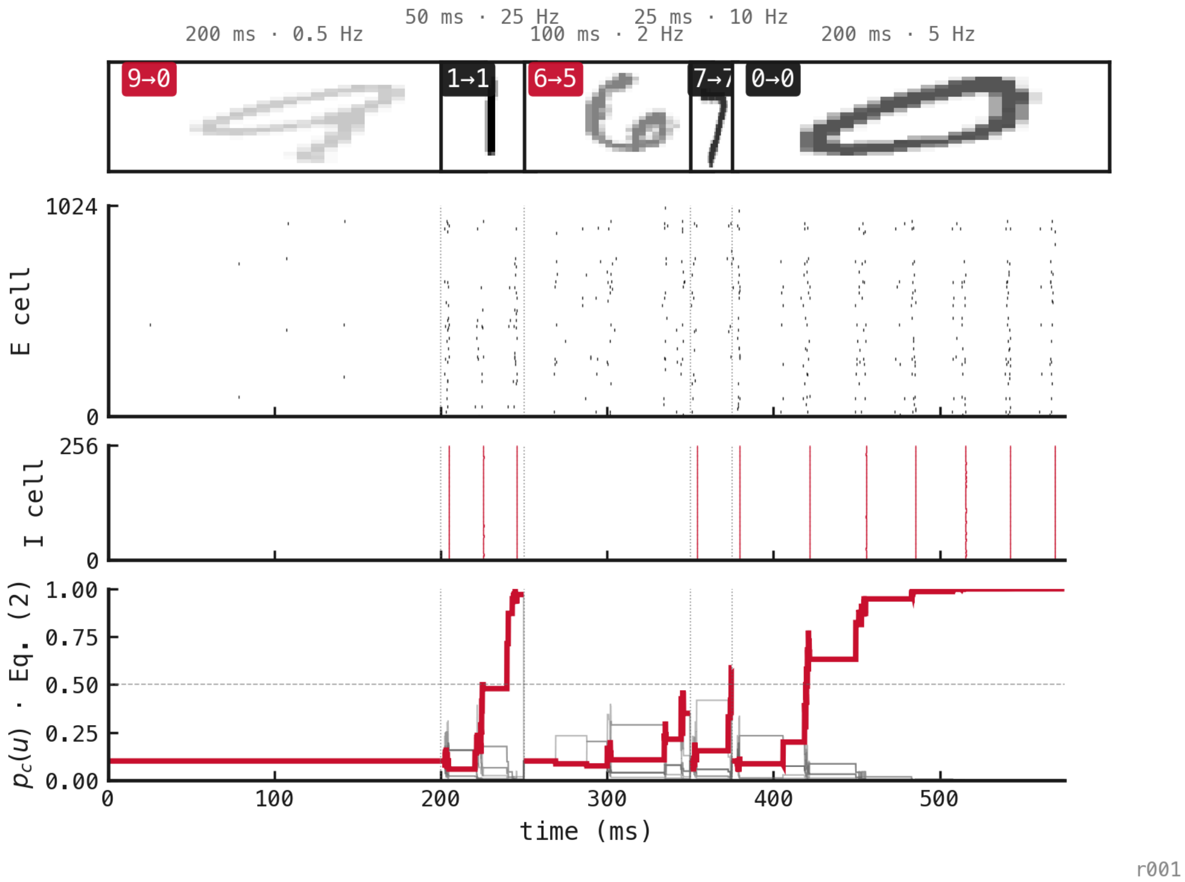


Figure 3: Successor to exp048 Figure 1, using the variable-rate-trained PING network and its native output-LIF spike-count readout. Presentation duration and maximum-pixel input rate vary between segments. Thumbnail opacity increases with encoding rate; badges show true label to prediction, with errors in red. The middle panels plot E- and I-cell spike rasters against time (ms). The lower panel plots $p_{c(u)}$ from Equation 2, with the true-class trace emphasized in red. The cumulative counts $z_{c(u)}$ from Equation 1 reset at each boundary while hidden PING state remains continuous.

The factorial summary separates the two manipulations. The left panel crosses presentation duration with maximum-pixel input rate. The right panel holds presentation and readout at 200 ms, giving the rate psychometric without a simultaneous change in integration time.

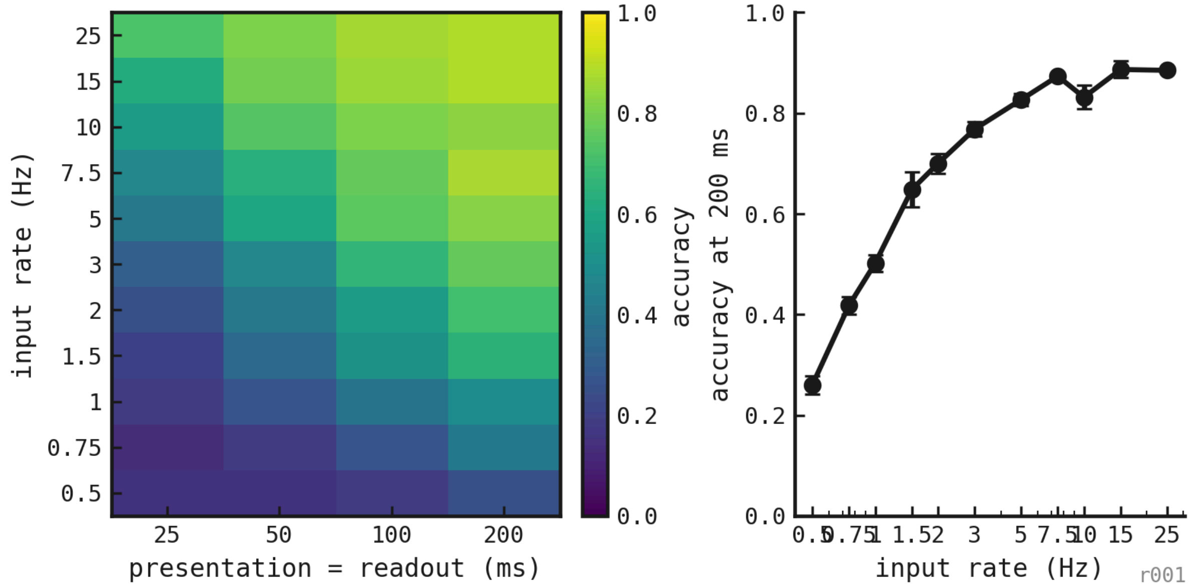


Figure 4: Accuracy across matched presentation/readout durations and input rates, with the 200 ms rate psychometric shown separately.

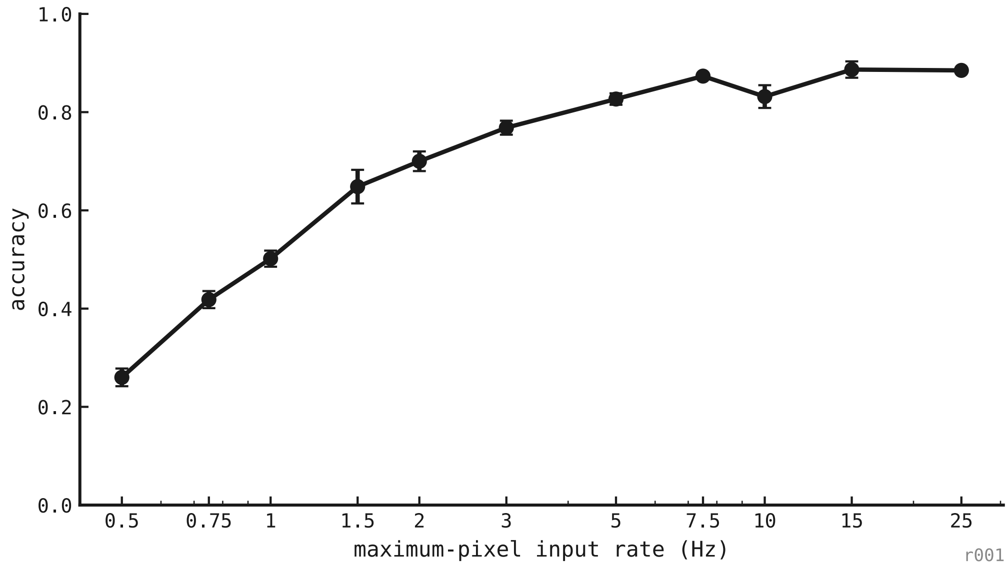


Figure 5: Input-rate psychometric at a fixed 200 ms presentation and spike-count window.

**3. Comparison with exp048**

Exp048 used a fixed-25-Hz training distribution and reconstructed a sliding `mem-mean` output. This experiment instead trains across the evaluated rates and uses the trained output-LIF spike counts directly. A numerical difference between the experiments therefore combines a change in training distribution with a change in readout; it is not an isolated comparison of either intervention.